Algebra

CHAPTER 10

Mathematical Reasoning 10.1–10.2

DPP 10.1 Mathematical Reasoning 10.1

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CHAPTER 10 Mathematical Reasoning

DPP 10.1
Mathematical Reasoning

Single Correct Answer Type

1. Which of the following is a logical statement?
(a) Are you going to Mumbai?
(b) Close the door.
(c) All odd integers are not prime numbers.
(d) What an excellent speech!

2. Which of the following is NOT a logical statement?
(a) Every square is a rectangle.
(b) 30 is divisible by 5.
(c) The sum of two odd integers is an even integer.
(d) Tomorrow is Sunday.

3. Consider the statement p : "Bengaluru is in India". Which of the following is NOT the negation of p?
(a) Bengaluru is not in India.
(b) It is false that Bengaluru is in India.
(c) It is not the case that Bengaluru is not in India.
(d) It is not the case that Bengaluru is in India.

4. The negation of the statement "3 is not a rational number" is
(a) 3 is a prime number.
(b) 3 is a real number.
(c) 2 is a real number.
(d) 3 is a number of type p/q, where $p, q \in Z$; $q \neq 0$.

5. Which of the following is NOT a component statement of the statement '30 is divisible by 2, 3, and 5'?
(a) 30 is divisible by 2. (b) 30 is divisible by 3.
(c) 30 is divisible by 5. (d) 30 is divisible 15.

6. Statement p : "Rakesh is in office"
Statement q : "Rakesh is at home".
Now, consider the following two statements:
S_1: $\sim (p \wedge q)$ is "Rakesh is not in office or not at home"
S_2: $\sim(p \vee q)$ is "Rakesh is neither in office nor at home".
Then,
(a) Only S_1 is true
(b) Only S_2 is true
(c) Both S_1 and S_2 are false
(d) Both S_1 and S_2 are true

7. If statement $(p \to q) \to r$ is false, then which of the following can't be truth values of p, q and r?
(a) T, T, F (b) F, F, F
(c) F, T, F (d) T, F, F

8. If statement p is true and statement q is false, then
S_1: $[\sim(p \leftrightarrow \sim q)] \to \sim q$ is true
S_2: $(\sim p \to q) \to (\sim q \to \sim p)$ is true
Then,
(a) Only S_1 is true
(b) Only S_2 is true
(c) Both S_1 and S_2 are false
(d) Both S_1 and S_2 are true

9. Negation of the statement $p \to (q \wedge r)$ is
(a) $\sim p \to \sim(q \wedge r)$ (b) $\sim p \vee (q \wedge r)$
(c) $(q \wedge r) \to p$ (d) $p \wedge (q \to \sim r)$

10. If $x = 20$ and $y = -4$, then $x \div y = -5$. The contrapositive of this statement is
(a) If $x \div y = -5$, then $x = 20$ and $y = -4$.

(b) If $x \div y \neq -5$, then neither $x \neq 20$ nor $y \neq -4$.
(c) If $x \div y \neq -5$, then $x \neq 20$ or $y \neq -4$.
(d) If $x \div y \neq -5$, then either $x \neq 20$ or $y = -4$.

11. $\sim(p \vee q) \vee (\sim p \wedge q)$ is logically equivalent to
(a) $\sim p$ (b) p
(c) q (d) $\sim q$

12. The statement $(p \to q) \to [(\sim p \to q) \to q]$ is
(a) a tautology (b) equivalent to $\sim p \to q$
(c) equivalent to $p \to \sim q$ (d) a fallacy

13. Statement $((\sim q) \wedge p) \vee (p \vee \sim p)$ is/equivalent to
(a) a tautology (b) a fallacy
(c) $\sim p \vee \sim q$ (d) $p \vee q$

14. Statement $[(p \leftrightarrow q) \wedge ((q \to r) \wedge r)] \to r$ is/equivalent to
(a) a fallacy (b) a tautology
(c) $\sim p \wedge \sim q$ (d) $p \vee q$

15. The statement $[p \wedge (p \to q)] \to q$ is
(a) a fallacy
(b) a tautology
(c) neither a fallacy nor a tautology
(d) none of these

16. If the compound statement $p \to (\sim p \vee q)$ is false, then the truth values of p and q are, respectively,
(a) T, T (b) T, F
(c) F, T (d) F, F

17. Consider the following two statements:
S_1: The negation of $p \to (p \vee \sim q)$ is a fallacy.
S_2: The negation of the statement $(p \wedge r) \to (r \vee q)$ is a tautology.
(a) Only S_1 is true
(b) Only S_2 is true
(c) Both S_1 and S_2 are false
(d) Both S_1 and S_2 are true

18. Consider the following statements:
p: Team India plays well.
q: Virat Kohli is the captain.
Now, which of the following is NOT true?
(a) The converse of $p \to q$ is "If Virat Kohli is the captain, then Team India plays well"
(b) The inverse of $p \to q$ is "If Team India does not play well, then Viral Kohli is not the captain"
(c) The contrapositive of $p \to q$ is "If Viral Kohli is not the captain, then Team India does not play well"
(d) None of these

19. The proposition $\sim(p \vee \sim q) \vee \sim(p \vee q)$ is logically equivalent to
(a) p (b) q
(c) $\sim p$ (d) $\sim q$

20. The negation of the statement "If a quadrilateral is a square, then it is a rhombus" is
(a) If a quadrilateral is not a square, then it is a rhombus.
(b) If a quadrilateral is a square, then it is not a rhombus.
(c) A quadrilateral is a square and it is not a rhombus.
(d) A quadrilateral is not a square and it is a rhombus.

21. The negation of the statement "Two lines intersect if and only if they have different slopes" is
(a) Two lines do not intersect and they have different slopes.
(b) Two lines intersect and they do not have different slopes.
(c) Two lines do not intersect and they do not have different slopes.
(d) Either two lines intersect and they have same slope or two lines do not intersect and they have different slopes.

Answer Key

Single Correct Answer Type

1. (c)	**2.** (d)	**3.** (c)	**4.** (d)	**5.** (d)	**6.** (d)	**7.** (d)	**8.** (a)	**9.** (d)	**10.** (c)
11. (a)	**12.** (a)	**13.** (a)	**14.** (b)	**15.** (b)	**16.** (b)	**17.** (a)	**18.** (d)	**19.** (c)	**20.** (c)
21. (d)									

CHAPTER 10

DPP 10.1

Single Correct Answer Type

1. (c)

2. (d)

3. (c)

4. (d) Negation of the given statement is "3 is a rational number" or "3 is number of type p/q, where $p, q \in Z$; $q \neq 0$"

5. (d)

6. (d)

7. (d) $(p \to q) \to r$ is false

$\therefore$ $(p \to q)$ is T and r is F

$\Rightarrow$ (p is T, q is T and r is F)

or (p is F, q is F and r is F) or (p is F, q is T and r is F)

8. (a) p is true and q is false

$\Rightarrow$ p is true and $\sim q$ is true

$\Rightarrow$ $p \leftrightarrow \sim q$ is true

$\Rightarrow$ $\sim (p \leftrightarrow \sim q)$ is false

$\Rightarrow$ $[\sim(p \leftrightarrow \sim q)] \to \sim q$ is true.

Also, $\sim p \to q$ is true and $\sim q \to \sim p$ is false

$\Rightarrow$ $(\sim p \to q) \to (\sim q \to \sim p)$ is false

9. (d) $\sim(p \to (q \wedge r)) \equiv \sim(\sim p \vee (q \wedge r))$

$\equiv p \wedge \sim (q \wedge r) \equiv p \wedge (\sim q \vee \sim r)$

$\equiv p \wedge (q \to \sim r)$

10. (c) Let $p : x = 20$, $q : y = -4$ and $r : x \div y = -5$.

Therefore, the given statement is $(p \wedge q) \to r$.

Its contrapositive is $\sim r \to \sim(p \wedge q)$

or $\sim r \to (\sim p \vee \sim q)$ or $x \div y \neq -5$ then $x \neq 20$ or $y \neq -4$

11. (a) Use Venn diagram.

12. (a) $(p \to q) \to [(\sim p \to q) \to q]$

$\equiv (\sim p \vee q) \to ((p \vee q) \to q)$

$\equiv \sim(\sim p \vee q) \vee (\sim(p \vee q) \vee q)$

$\equiv$ tautology (using Venn diagram)

13. (a) $((\sim q) \wedge p) \vee (p \vee \sim p) \equiv ((\sim q) \wedge p) \vee t \equiv t$

14. (b)

p	q	r	$p \leftrightarrow q$	$q \to r$	$(q \to r) \wedge r$	$[(p \leftrightarrow q) \wedge ((q \to r) \wedge r)]$	$[(p \leftrightarrow q) \wedge ((q \to r) \wedge r)] \to r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	F	T
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	T	F	F	F	F	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	F	F	T

15. (b) $[p \wedge (p \to q)] \to q$

$= \sim [p \wedge (p \to q)] \vee q$

$\equiv \sim p \vee (\sim (\sim p \vee q)) \vee q$

$\equiv \sim p \vee (p \wedge \sim q) \vee q$

$\equiv (\sim p \vee p) \wedge (\sim p \vee \sim q) \vee q$

$\equiv t \wedge (\sim p \vee \sim q) \vee q$

$\equiv (\sim p \vee \sim q) \vee q$

$\equiv \sim p \vee (\sim q \vee q)$

$\equiv \sim p \vee t = t$

16. (b) $p \to (\sim p \vee q)$ is false

$\Rightarrow$ p is true and $(\sim p \vee q)$ is false

$\Rightarrow$ p is true and q is false

17. (a) $p \to (p \vee \sim q)$

$\equiv \sim p \vee (p \vee \sim q)$

$\equiv (\sim p \vee p) \vee \sim q$

$\equiv t \vee \sim q$

$\equiv t$

So, negation of given statement is a fallacy

Now, $(p \wedge r) \to (r \vee q)$

$\equiv \sim (p \wedge r) \vee (r \vee q)$

$\equiv (\sim p \vee \sim r) \vee (r \vee q)$

$\equiv (\sim p \vee q) \vee (\sim r \vee r)$

$\equiv (\sim p \vee q) \vee t$

$\equiv t$

So, negation of given statement is fallacy.

18. (d) Converse of $p \to q$ is $q \to p$.

Inverse of $p \to q$ is $\sim p \to \sim q$.

Contrapositive of $p \to q$ is $\sim q \to \sim p$.

19. (c) $\sim(p \vee \sim q) \vee \sim(p \vee q)$

$\equiv \sim [(p \vee \sim q) \wedge (p \vee q)]$

$\equiv \sim [(p \vee (\sim q \wedge q)]$

$\equiv \sim [p \vee f] \equiv \sim p$

20. (c) Let p and q be the statements as given below.

p : a quadrilateral is a square.

q : a quadrilateral is a rhombus.

The given statement is $p \to q$.

$\therefore$ $\sim(p \to q) \equiv \sim(\sim p \vee q) \equiv p \wedge \sim q$

Therefore, the negation of the given statement is "A quadrilateral is a square and it is not a rhombus"

21. (d) Let p and q be the statements as given below.

p : Two lines intersect.

q : They have different slopes.

Then, the given statement is $p \leftrightarrow q$.

Now, $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$

Therefore, $\sim(p \leftrightarrow q)$:

Either two lines intersect and they have same slope or two lines do not intersect and they have different slopes.